Bee Mortality Analysis from Bijensterfte Survey Data 2024

Presented by: Ming Zhang
9.12.2025

198

Responses

Beefoundation

meters maken voor bijen, hommels en vlinders

Section 1 of 8

Honingbijen sterfte 24-25

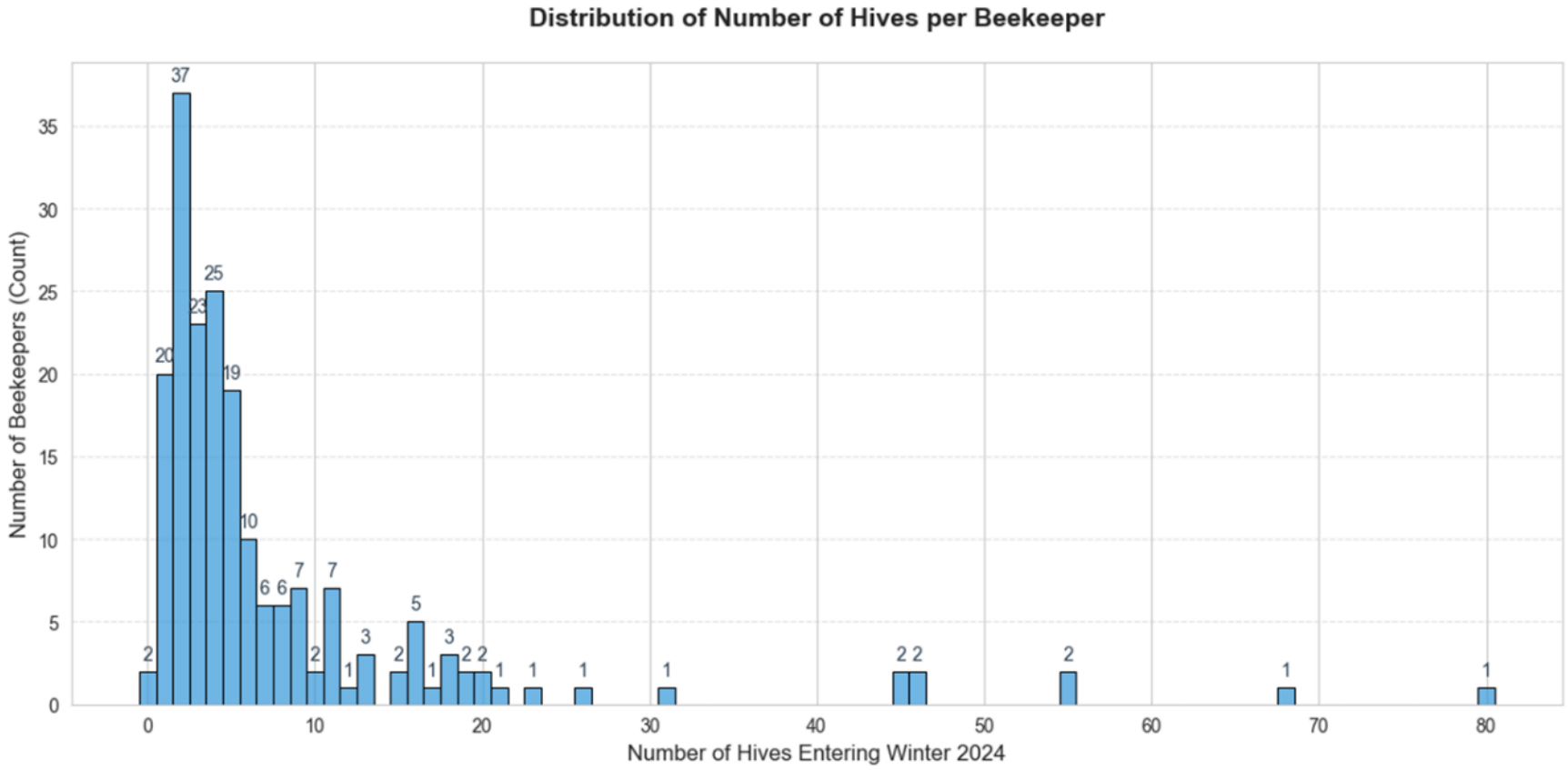
Beste imker,
Voor het derde jaar horen we van imkers dat de afgelopen zomer, herfst en winter weer veel bijenvolken gestorven zijn. Dit verlies is voor ons allen zeer pijnlijk. Bovendien: honingbijen zijn een wezenlijk deel van onze leefomgeving. Zij spiegelen de gezondheid van de omgeving waarvan wij allemaal deel uitmaken. Wat is er de afgelopen jaren aan het veranderen waardoor de sterfte zulke grote proporties aanneemt?

Waarom deze enquête?
Met deze enquête proberen we de vinger achter dit sterven te krijgen. Is het blootstelling aan pesticiden, klimaatverandering, de bedrijfsmethode van de imker, ziekteverwekkers of voedselaanbod? Of is het een combinatie van factoren? En het allerbelangrijkste: wat kunnen we doen om deze sterfte te keren?

Er lijkt een grillig sterftepatroon te zijn: op de ene plek zijn alle volken dood, op de andere is er niets aan de hand. Dit maakt dat het belangrijk is om gegevens te verzamelen van veel verschillende imkers dicht bij elkaar in de buurt. Deel daarom deze enquête met uw buurimkers. Als we samen meer gegevens verzamelen, krijgen we samen een beter inzicht.

Descriptive View of the Dataset

- The distribution of number of hives among respondents is highly right-skewed. The vast majority of the study population consists of small-scale, hobbyist beekeepers.
- At the beginning of the 2024 winter season, over 80% of respondents owns 10 or fewer hives. Therefore, our data to a large extent represents the average beekeeper's experience.



Professional
9
13.7%

Hobbyist
189
25.7%

Mortality Rate

Data Conversion: from beekeeper to hives

- The original survey contained beekeeper-level records, where one entry may include multiple hives. To analyze the possible effects at the hive level, this dataset was converted into hive-level entries (where applicable).

	A	B	C	D	E	F	G	H
1	**id**	**Postcode cijfers**	**Postcode Letters**	**GPS_x**	**GPS_y**	**Province**	**2024_spring**	**2024_winter**
2	id	Postcode cijfers	Postcode Letters	Vul hier je GPS locatie in		In welke provincie staan je bijen?	Met hoeveel volken begon je dit bijenjaar (voorjaar 2024?)	Hoeveel volken heb je ingewinterd? (najaar 2024)
4	1	10XX	XX			Utrecht	8	12
5	2	73XX	XX			Gelderland	2	2
6	3			52.XXXXX	6.XXXXX	Overijssel	3	4
7	4	30XX	XX			Zuid-Holland	4	4
8	5	70XX	XX			Gelderland	6	4
9	6			52.XXXXX	5.XXXXX	Noord-Holland	1	2

Beekeeper-level entries from the survey

	A	B	G	H	I	J	K	L	N	O
1	hive_id	hive_lifecycle_status	id	Postcode cijfe	Postcoc	GPS_y	GPS_x	Province	Death_rate_winte	Growth_rate_24_25
2	2_hive_1	Survived Winter 24	2	73XX	XX			Gelderland	0.5	-0.
3	2_hive_2	Died During Winter 24	2	73XX	XX			Gelderland	0.5	-0.
4	3_hive_1	Survived Winter 24	3			52.XXXXX	6.XXXXX	Overijssel	0.25	
5	3_hive_2	Survived Winter 24	3			52.XXXXX	6.XXXXX	Overijssel	0.25	
6	3_hive_3	Survived Winter 24	3			52.XXXXX	6.XXXXX	Overijssel	0.25	
7	3_hive_4	Died During Winter 24	3			52.XXXXX	6.XXXXX	Overijssel	0.25	
8	4_hive_1	Survived Winter 24	4	31XX	XX			Zuid-Holland	0.5	-0.
9	4_hive_2	Survived Winter 24	4	31XX	XX			Zuid-Holland	0.5	-0.
10	4_hive_3	Died During Winter 24	4	31XX	XX			Zuid-Holland	0.5	-0.
11	4_hive_4	Died During Winter 24	4	31XX	XX			Zuid-Holland	0.5	-0.
12	5_hive_1	Survived Winter 24	5	73XX	XX			Gelderland	0.25	-0.
13	5_hive_2	Survived Winter 24	5	73XX	XX			Gelderland	0.25	-0.
14	5_hive_3	Survived Winter 24	5	73XX	XX			Gelderland	0.25	-0.
15	5_hive_4	Died During Winter 24	5	73XX	XX			Gelderland	0.25	-0.
16	5_hive_5	Lost Before Winter 24	5	73XX	XX			Gelderland	0.25	-0.

Hive-level entries

Data Conversion

- The first step is to identify the total number of hives each beekeepers have had.
  - Identify the number of hives that belongs to each beekeeper by identifying the largest number from *2024_spring*, *2024_winter*, and *2025_spring*. If a beekeeper reported n hives at a location, create n rows in Excel, each representing one hive.
- Add a few columns to describe the attributes of each hives:
  - *hive_id*: the unique identifier of a hive. In total, there are 2566 unique hives.
  - *hive_lifecycle_status*: whether the hive is lost before winter 2024, died during winter 2024, or survived winter 2024.

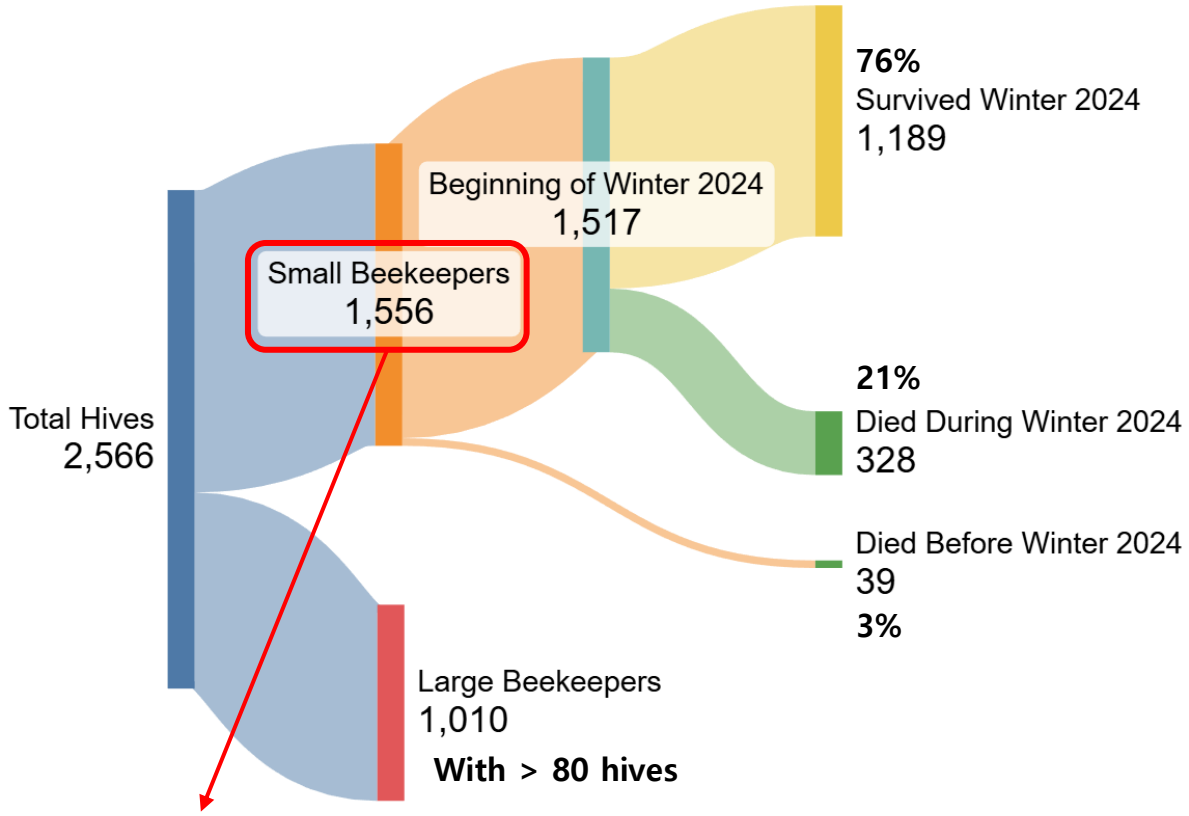
Beekeeper -level entries from the survey



Hive-level entries

Descriptive View of the Dataset

- The data cleansing process found that one beekeeper (id = 1) didn't give the number of hives in 2025 spring, and two beekeepers (id = 23 & 51) has a growth of hives during the winter. These three beekeepers are omitted from the dataset immediately.
- The dataset of the beehives continues with the data from 195 beekeepers. Altogether, they reported 2566 hives.
- There are two beekeepers who possess a substantial number of hives: id = 108 starts with 500 hives in spring 2024, and id = 120 starts with 80 hives in spring 2024.

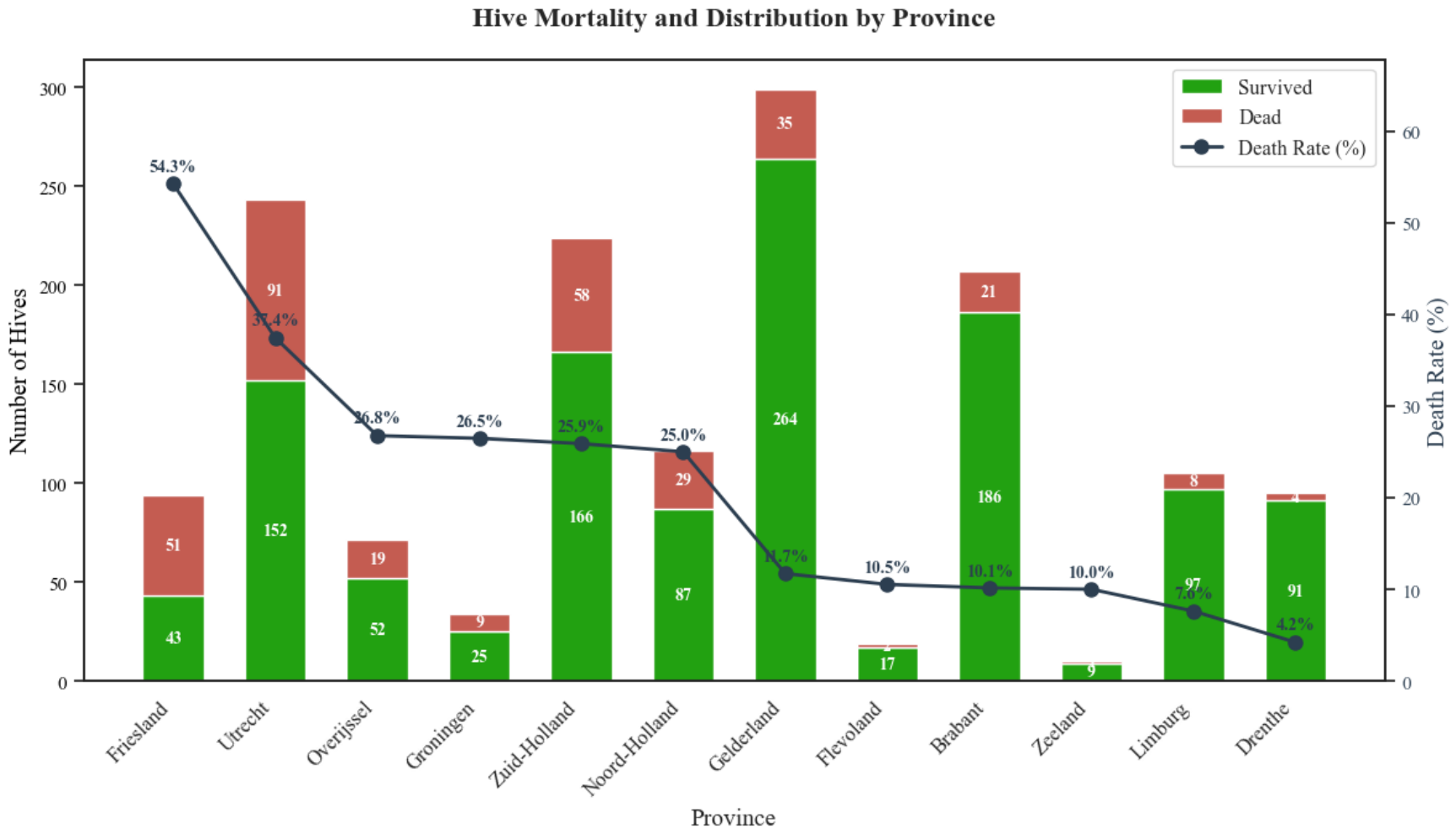


The following analysis focuses on the 1556 hives from small beekeepers.

24%
Mortality Rate

Descriptive View of the Dataset

- While mortality rates show high variance across provinces, a previous study by the Bee Foundation indicates that results are likely representative only for provinces with more than 300 hives.
- In this survey, while no single province met the 300-hive benchmark for robust statistical inference, this visualization still reveals province-level heterogeneity and serves as a valuable exploratory tool for identifying potential high-risk regions.



Possible Factors

Beekeeper level



Observation and subjective estimation

- Honey flow during summer
- Honey flow during autumn
- Development in spring and summer
- Observed diseases (with a focus on varroa)

Hive level



Systemic action and attribute of bees

- Honey use during winter
- Beekeeping method
- Varroa Treatment

Statistical method: unweighted

Statistical method: weighted

On the Beekeeper Level

This treats every beekeeper equally. We chose to present these results because they best reflect the reality for the majority of the beekeepers: the small-to-medium scale beekeepers.

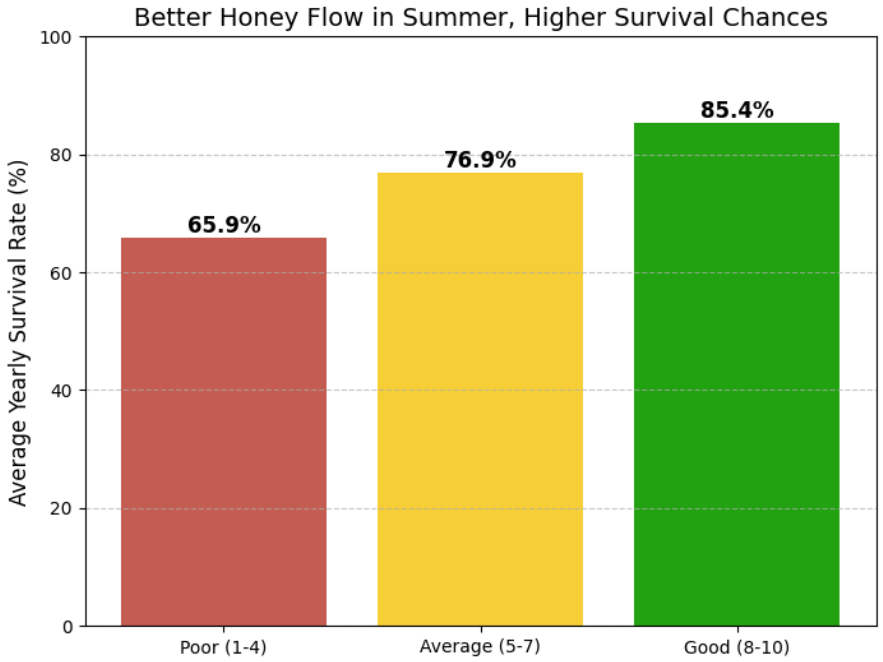
Unweighted Analysis

- Honey flow during summer
- Honey flow during autumn
- Development in spring and summer
- Observed diseases (with a focus on varroa)

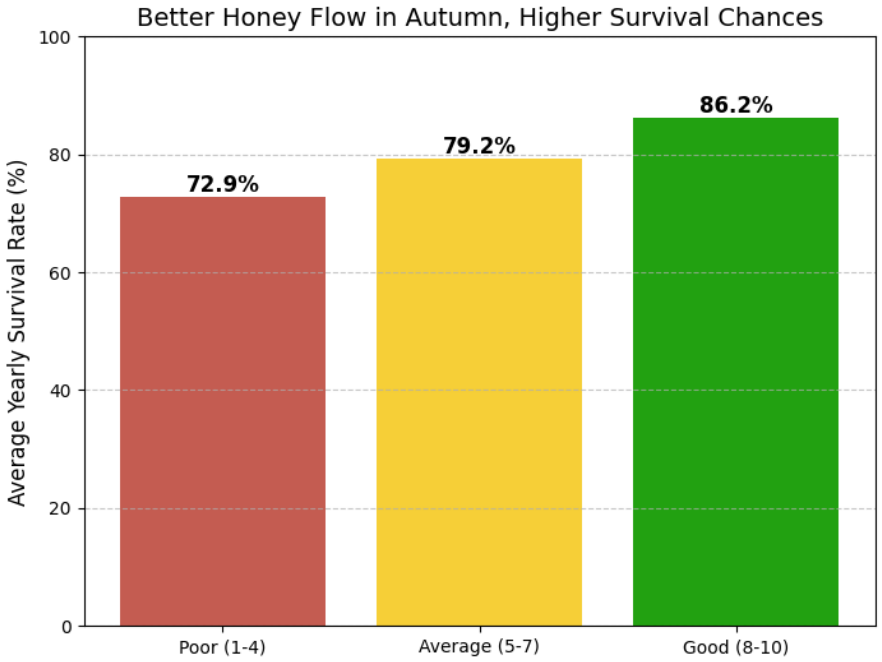


Honey Flow

- Honey flow is self-reported by beekeepers. On a scale of 1 to 10, beekeepers are asked to rate their honey flow in summer (June and July) and autumn (September and October).
- Visual trends suggest that better honey flow in both seasons is linked to higher survival. Statistical tests (Spearman Correlation) also confirm that honey flow in both summer and autumn is a significant predictor for survival rate. Better honey flow correlates with higher survival chances.
    - We used an unweighted Spearman Correlation to analyze these trends. This method treats each beekeeper equally, regardless of whether they manage 1 hive or 50 hives.



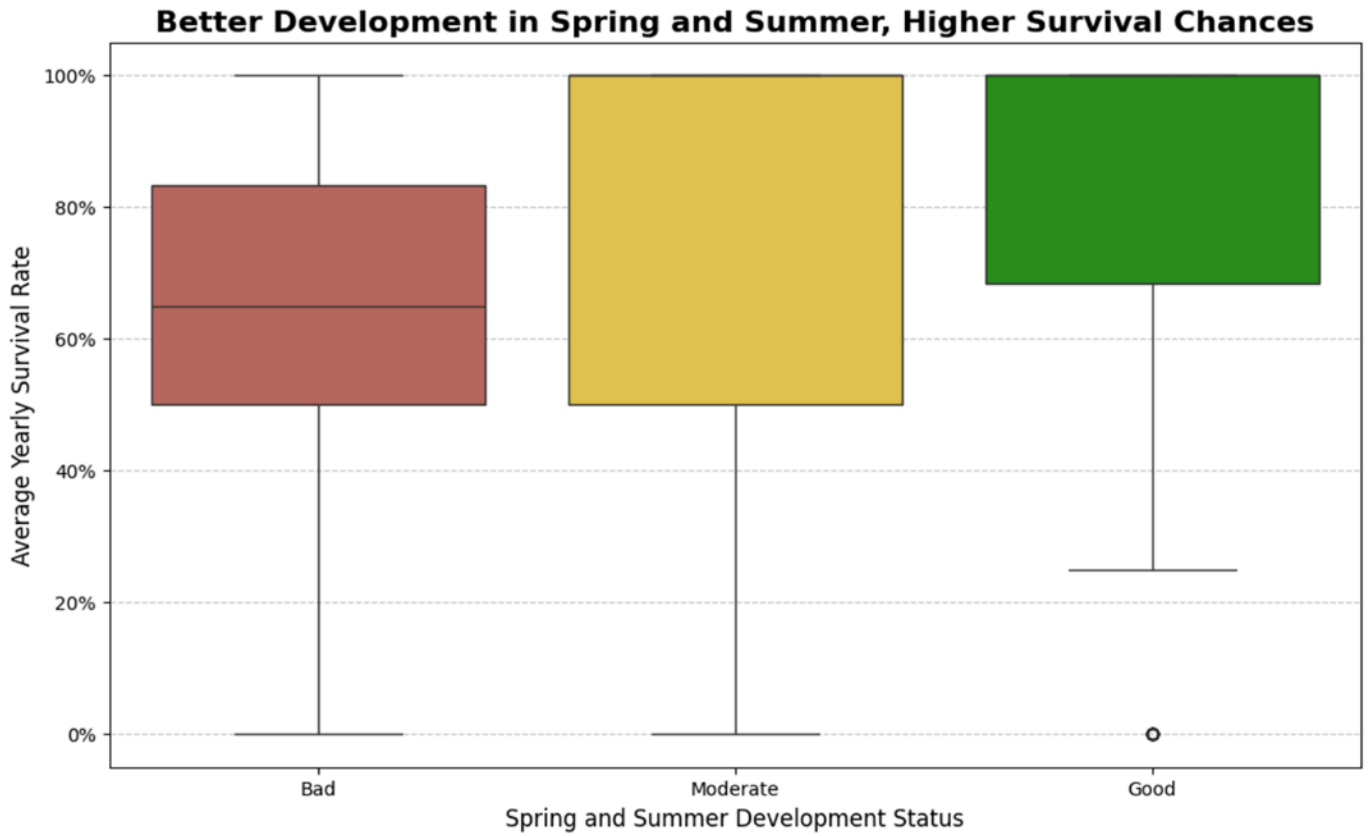
Spearman Correlation: $p<0.05$, $r = -0.20$.



Spearman Correlation: $p<0.05$, $r = -0.21$.

Development of Hives in Spring and Summer

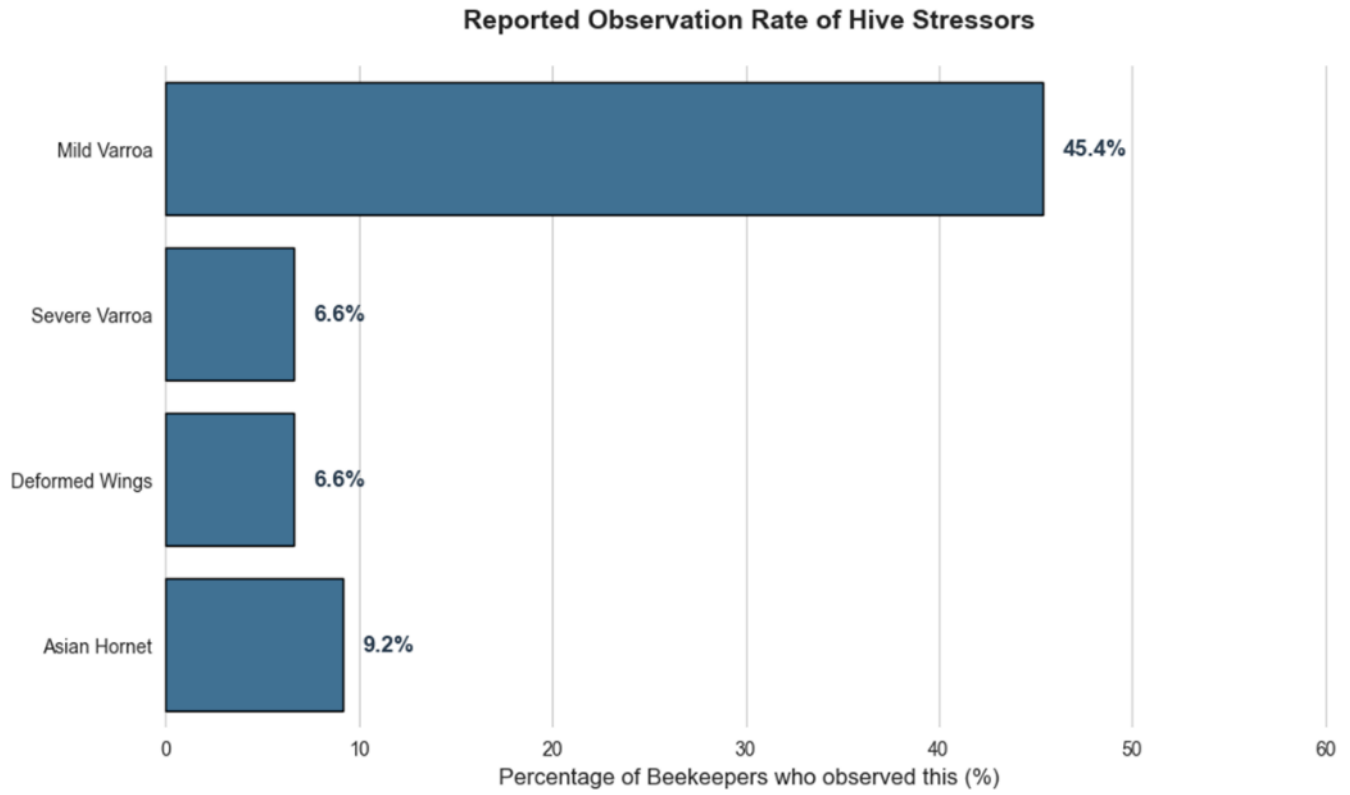
- Development of hives in spring and summer is self-reported by beekeepers by the question 'Hoe was de ontwikkeling van de volken tijdens het voorjaar en zomer 2024?'. They can choose from three options: Overwegend slecht, Overwegend matig, or Overwegend goed.
- Visual trends suggest that better development is linked to higher survival rate. Statistical tests (Kruskal-Wallis test) also confirm that this is a significant predictor for death rate. Better development in spring and summer correlates with higher survival chances.
    - We used an unweighted Kruskal-Wallis test. This method treats each beekeeper equally, regardless of whether they manage 1 hive or 50 hives.



Kruskal-Wallis test: $p < 0.05$.

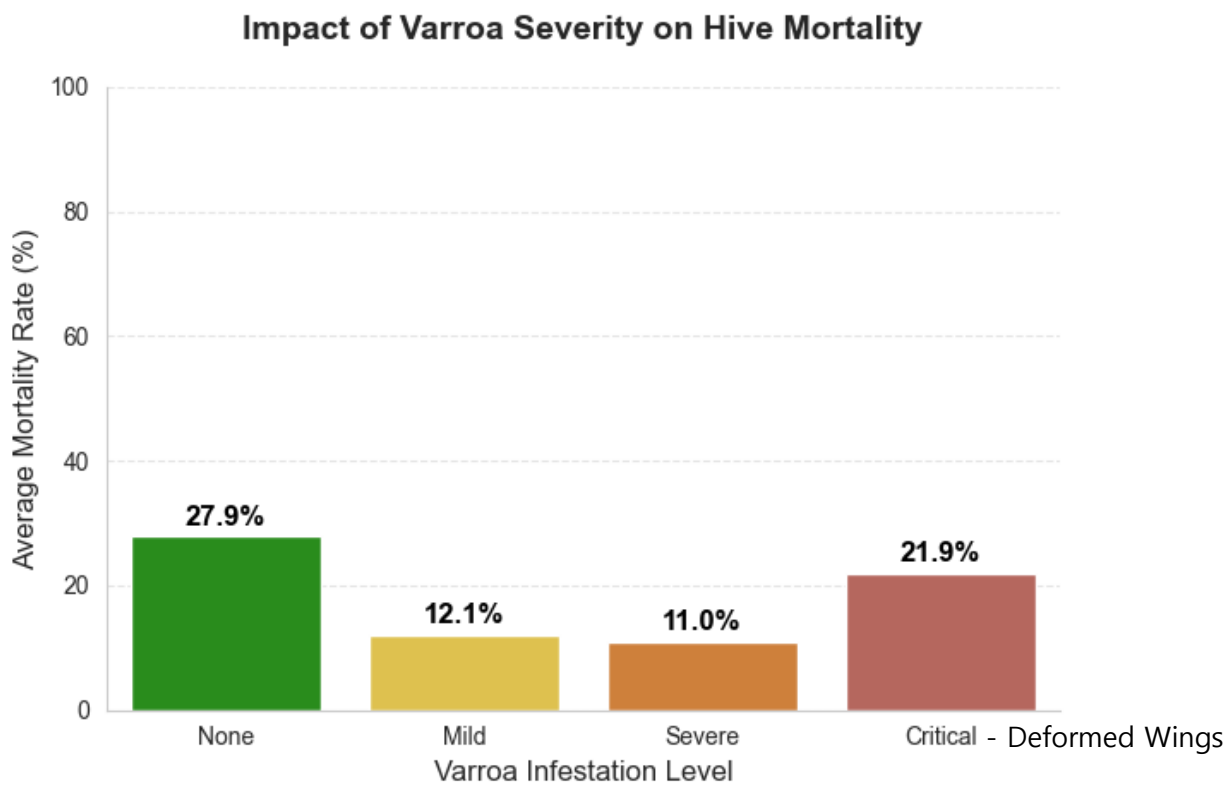
Varroa Severity

- Beekeepers are also asked to self-report diseases and other stressors they observed. As some of the diseases are not frequent, our analysis wants to focus on three key stressors that might contribute to bee mortality: varroa, deformed wings, and Asian hornet.
- Deformed wings can also be seen as as indicator of varroa. It often symbolizes a severe infection of varroa.
- Varroa is very common. 52% of the beekeepers reported seeing varroa, but the severity varies. (45.4% reported mild varroa, 6.6% reported severe varroa).
- Deformed wings can be seen as an indicator of varroa. 6.6% of beekeepers reported observing deformed wings.
- About 10% of the beekeepers reported Asian hornet.



The Invisibility Trap: Why Reporting "No Varroa" is Risky

- Initially, we saw something strange: Beekeepers who reported 'No Varroa' suffered significantly higher losses (28%) than those who reported mild (12%) or severe (11%) varroa infection.
- Why?
- Two hypothesis:
  - The difference in hive numbers create the difference, i.e. beekeepers with a large number of hives (which has a relatively stable death rate around the average rate) might be aggregated in the group of reporting mild or severe varroa, therefore skewed the data.
  - Self-reporting 'No Varroa' might mean 'lack of monitoring', which lead to 'lack of treatment' that causes death.

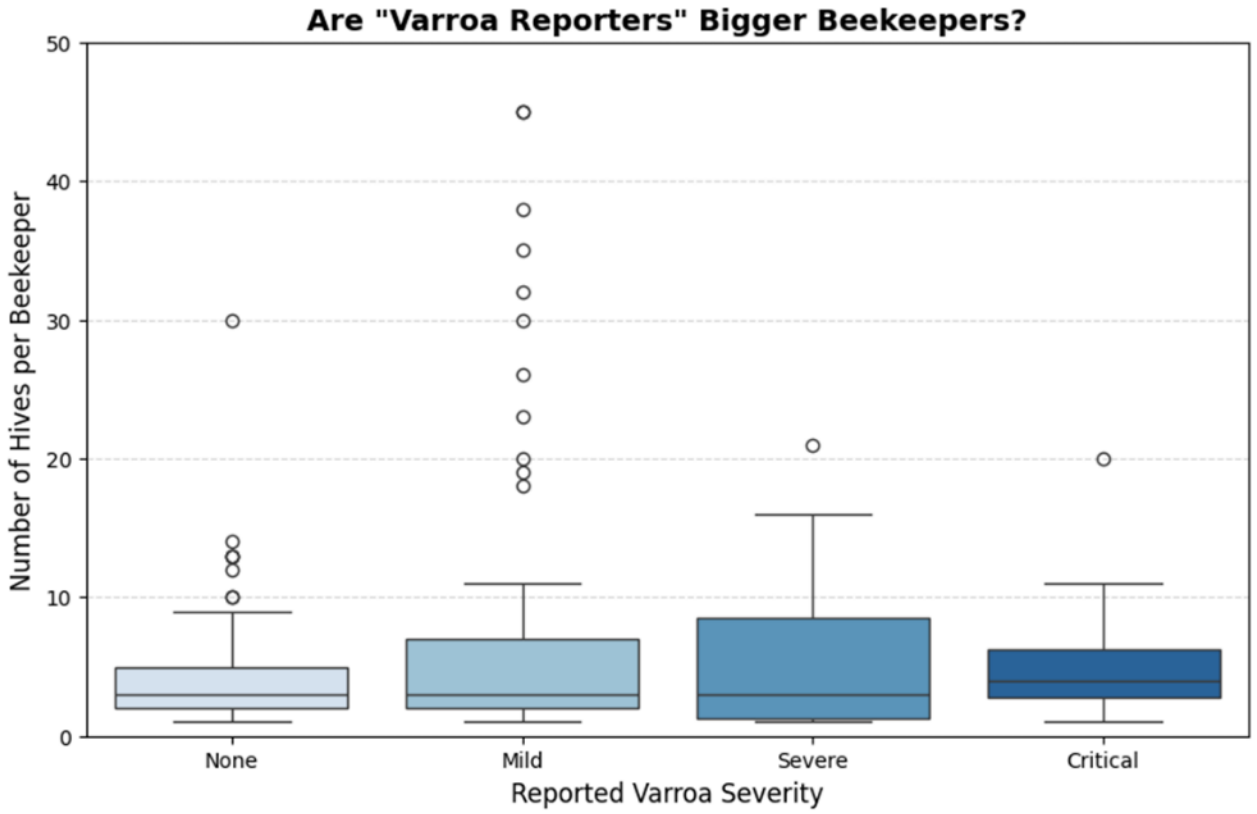


The Invisibility Trap: Why Reporting "No Varroa" is Risky

- Hypothesis 1:
  - The difference in hive numbers create the difference, namely beekeepers with a large number of hives might be aggregated in the group of reporting mild or severe varroa, therefore skewed the data (Because of the bigger amount of hives they have, their hive's death rate is relatively lower and stays around the average rate).

- Regardless of the number of hive a beekeeper owns, there is **no significant difference** between those who report varroa and those who don't.

	count	mean	median	max	min
Varroa_severity					
None	89	4.820225	3.0	64.0	1.0
Disease_varroa_Mild	73	7.726027	3.0	45.0	1.0
Disease_varroa_Severe	10	6.200000	3.0	21.0	1.0
Disease_deformed_wings	12	5.666667	4.0	20.0	1.0

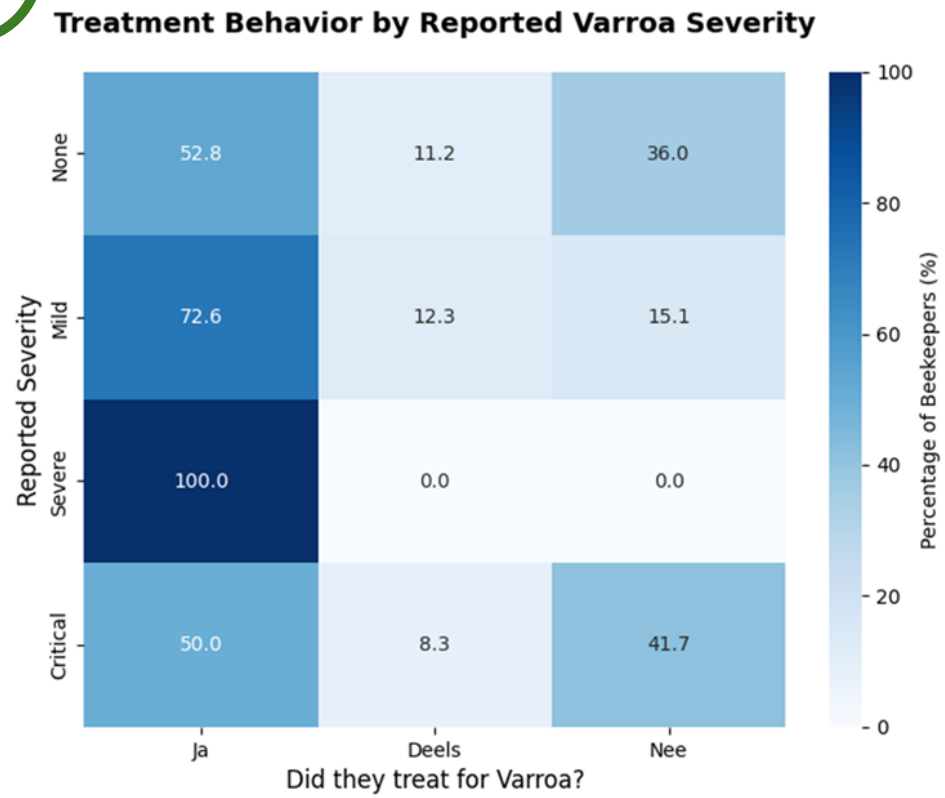


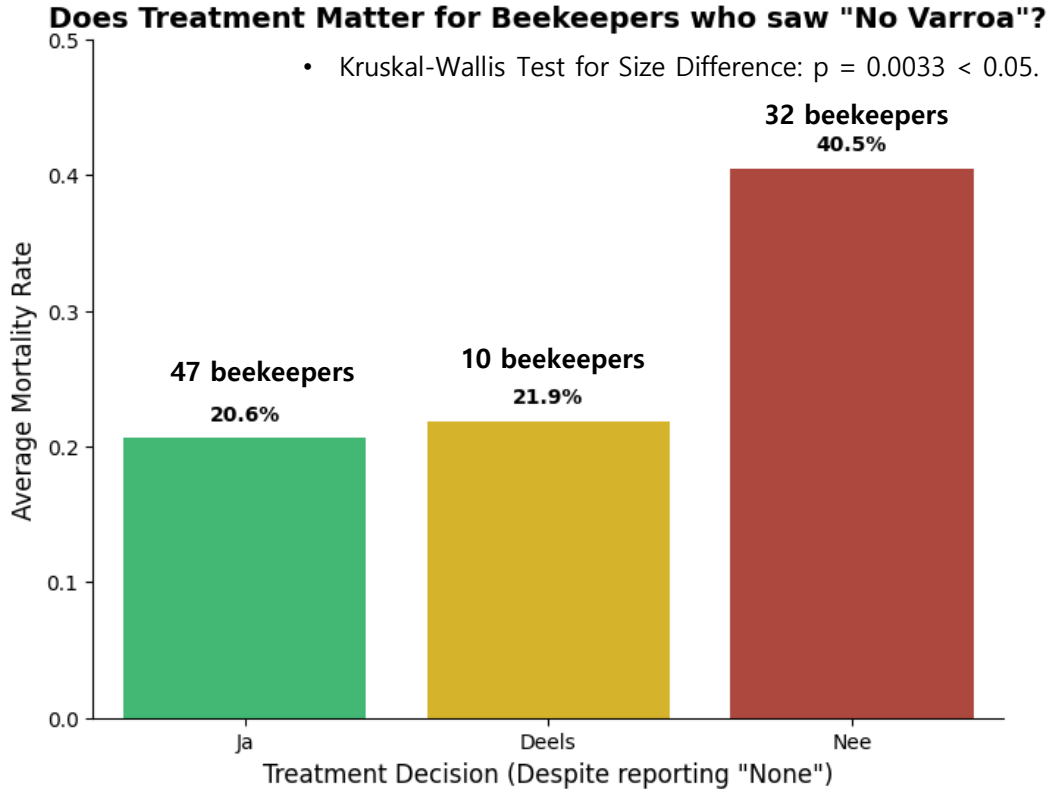
Kruskal-Wallis Test: $p = 0.2884 > 0.05$

The Invisibility Trap: Why Reporting "No Varroa" is Risky

- Hypothesis 2:
  - Self-reporting 'No Varroa' might mean 'lack of monitoring', which lead to 'lack of treatment' that causes death.

- No observed varroa leads to less treatment. However, **sometimes varroa are present, but undetected.** Providing varroa treatment significantly reduced mortality even when no varroa was observed.
- Not treating varroa is highly risky. Observing varroa is not a deathly sign, on the contrary, it means the beekeeper is monitoring, and more importantly, likely to be treating against varroa.





On the Hive Level

This focuses on the total bee population. The results here were mixed, largely because of a few larger hive clusters that behave differently from the norm. The more hives a beekeeper has, the more effects it has on the result.

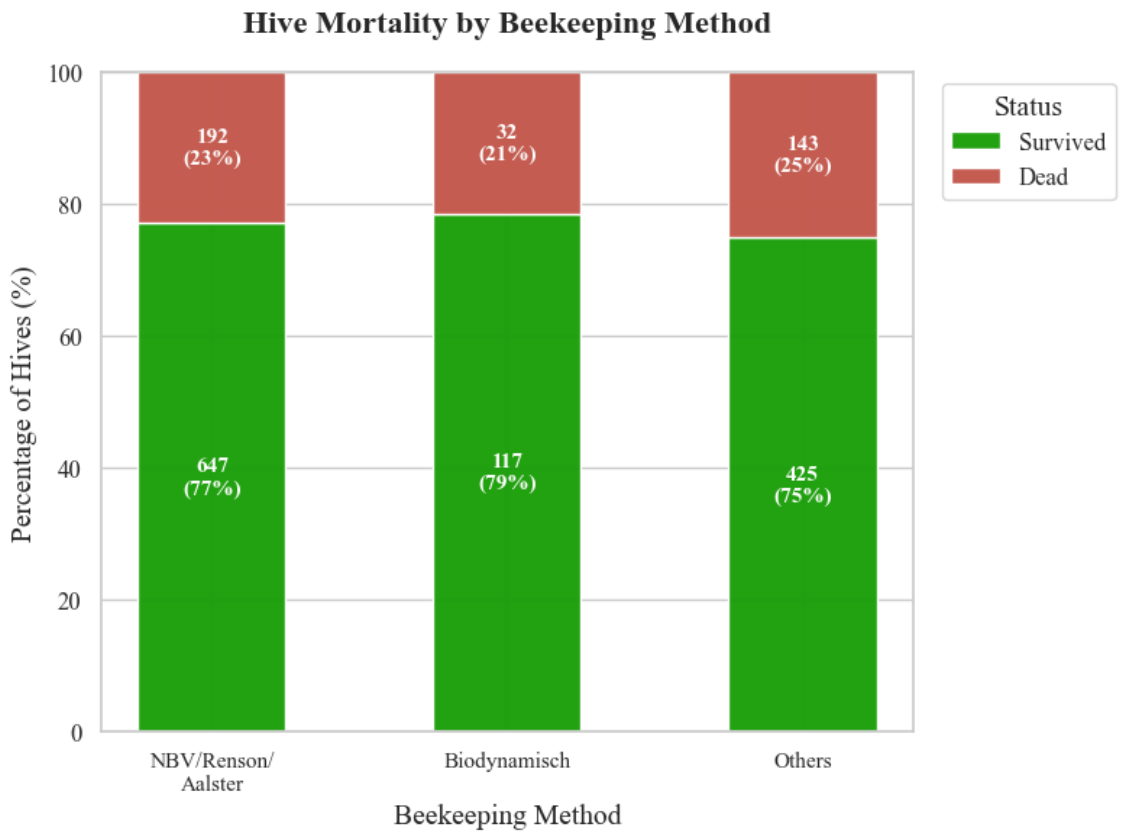
Weighted Analysis

- Beekeeping method
- Honey use during winter
- Varroa Treatment



Beekeeping Method

- Beekeeping method is grouped into three categories: group A with NBV/Renson/Aalster, group B with Biodynamisch, and group C with mixed methods or beekeepers who are unsure of their method.
- Different beekeeping methods show a comparatively balanced death rate. The NBV/Renson/Aalster method and Biodynamisch method respectively has a death rate of 23% and 21%.

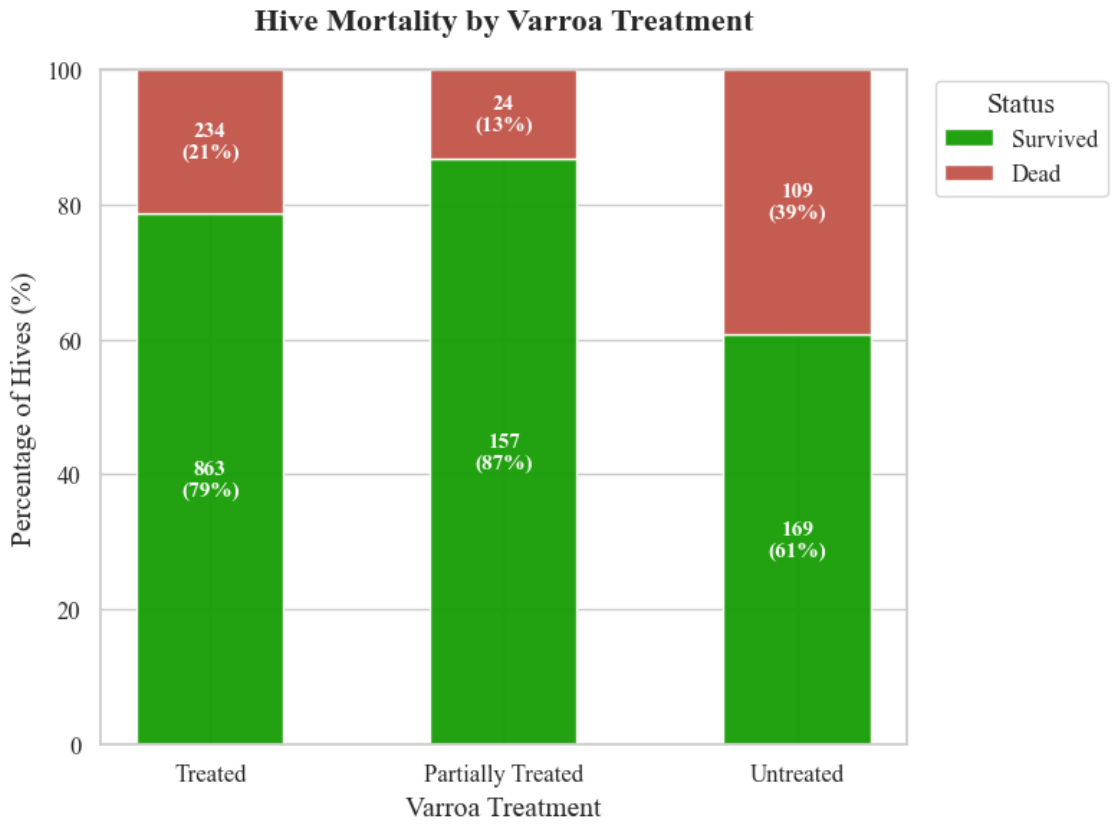


	NBV/Renson /Aalster		Biodynamisch		Others		Total	
Sum of hives	839		149		568		1556	
Dead	192	23%	32	21%	143	25%	367	24%
Survived	647	77%	117	79%	425	75%	1189	76%

Chi-Square Test: $p > 0.05$.

Varroa Treatment

- The majority of hives are treated against varroa, be it fully treated or partially treated. Only 17.86% are fully untreated against varroa.
- Hives untreated against varroa has a higher death rate (39% compared with 21% for those fully treated). Therefore, giving varroa treatment is proved to be an effective method to lower hive mortality.
- Surprisingly, hives receiving partial treatment showed the lowest mortality rate (13%).
    - The reason is unclear.



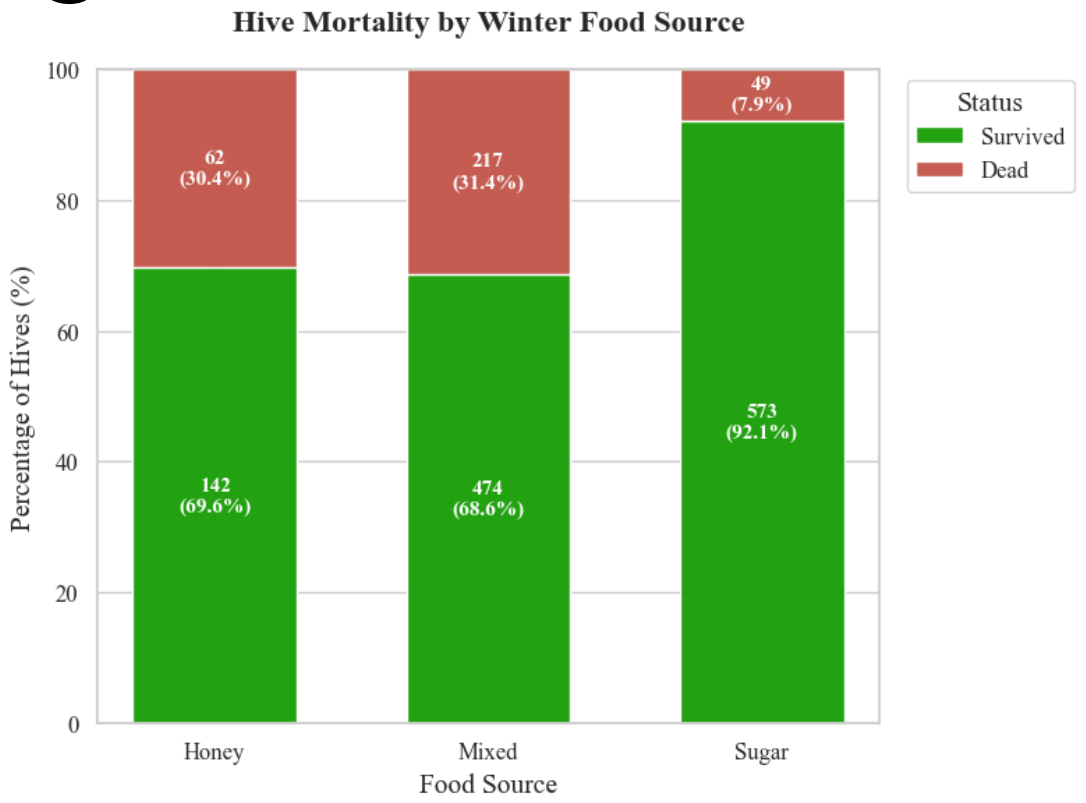
	Treated		Partially Treated*		Untreated		Total	
Sum of hives	1097		181		278		1556	
Dead	234	21%	24	13%	109	39%	367	24%
Survived	863	79%	157	87%	169	61%	1189	76%

Chi-Square Test: $p < 0.05$.

*'Partially treated' refers to being treated against varroa only in summer or only in winter.

Honey/Sugar Use during Winter

- The use of honey during winter contributes to a higher death rate. Hives that doesn't have honey in their winter food has a significantly lower death rate.



*Only including the 1517 hives that enters the winter of 2024.

	Honey		Mixed		Sugar		Total	
Sum of hives	204		691		622		1517*	
Dead (during winter)	62	30%	217	31%	49	8%	328	22%
Survived (during winter)	142	70%	474	69%	573	92%	1189	78%

Chi-Square Test (3 groups): $p < 0.05$.
Chi-Square Test (pairwise): $p = 0.85 > 0.05$ for honey vs mixed,
$p < 0.05$ for honey vs sugar, and for mixed vs sugar.

Does feeding sugar increases the risk of varroa?

- The analysis shows a strong negative association between sugar feeding and varroa severity. Hives overwintered on sugar consistently showed lower infestation levels compared to those on honey.
- However, this is likely due to management practices: Beekeepers who feed sugar are more likely to engage in active Varroa treatment regimes, whereas those who strictly feed honey may adopt 'natural' beekeeping methods with limited intervention, resulting in higher varroa presence.

Varroa Severity	Varroa treatment / Winter honey: Ja – Honey	Ja – Sugar	Deels – Honey	Deels – Sugar	Nee – Honey	Nee – Sugar
None	32.72% 195	42.32% 212	50.00% 37	5.61% 6	42.34% 105	13.33% 4
Mild	48.32% 288	53.29% 267	41.89% 31	94.39% 101	47.18% 117	56.67% 17
Severe	13.59% 81	0.80% 4				
Critical (Deformed Wings)	5.37% 32	3.59% 18	8.11% 6		10.48% 26	30.00% 9

Chi-Square Test (Group 'Ja' and Group 'Nee'): $p < 0.05$.

Multi-Variable Analysis

- While single-variable analysis is a crucial first step for identifying potential risk factors, it examines each variable in isolation.
- However, factors might overlap. A multi-factor model is necessary to filter out this noise and find the true drivers of survival.

If **all the beekeepers** change from feeding the bees honey during winter to feeding sugar, what is the difference in death rate?

GEE: population-averaged model

Statistical Model: Generalized Estimating Equations (GEE)

- GEE is a population-averaged model. It estimates the average effect across the entire population. It understands all the hives from one beekeeper as a cluster.
- Given the goal is to analyze the possible factors for hive mortality, using GEE is highly appropriate.

Variables

Dependent Variable (Y): Hive status (Binary, 1 = Dead, 0 = Survived)
Key Independent Variables (X):
Development in spring and summer (Categorical: Good, Moderate, Bad). *Reference level: Bad*
Honeyflow during summer (Ordinal: Poor, Average, Good). *Reference level: Poor*
Honeyflow during autumn (Ordinal: Poor, Average, Good). *Reference level: Poor*
Varroa severity (Categorical: None, Mild Varroa, Severe Varroa, Deformed Wings) *Reference level: None.*
Honey use during winter (Categorical: Honey, Mixed, Sugar). *Reference level: Sugar.*
Varroa treatment (Categorical: Yes, Either Summer or Winter, None). *Reference level: Yes.*
Beekeeping method (Categorical: Controlled, Natural, Others/Don't Know) *Reference level: Controlled.*

As a beekeeper, if I change from feeding the bees honey during winter to feeding sugar, what is the difference in the death rate of **my hive**?

other subject-specific model

Multi-Variable Analysis

- Autumn honey flow also shows a significant correlation with mortality, particularly comparing the ones with a 'poor' rating with those with a 'good' rating. The ones with an 'average' rating doesn't behave significantly different than the 'good' ones.
  - Compared with hives with 'good' honey flow during autumn, the likelihood to die for hives with 'poor' honey flow is 2.2 times (e^0.7746) higher.
- Honey/sugar use during winter revealed a powerful effect for mortality. It matches with our previous single variable analysis: that hives being fed on their own honey has a higher chance of dying.
  - Compared with those fed fully on sugar, the likelihood hives that are fed partly sugar partly honey is 2.8 (e^1.0213) more times likely to die, and for hives that are fed completely on honey it's 3.9 (e^1.3697) times higher.
- Self reported mild varroa has a negative correlation with mortality. This matches with our finding in the previous single variable analysis: not observing varroa does not necessarily mean no varroa.
  - If observed mild varroa, the likelihood to die is only 60% of no observed varroa.
- Other variables doesn't show statistical significance on the level of $p<0.05$.
  - Including: hive development in spring and summer, honey flow in summer, Severe Varroa, Deformed Wings, Varroa Treatment, Beekeeping method.

GEE Regression Results

Dep. Variable:	Hive_Status	No. Observations:	1516
Model:	GEE	No. clusters:	187
Method:	Generalized Estimating Equations	Min. cluster size:	1
		Max. cluster size:	80
Family:	Binomial	Mean cluster size:	8.1
Dependence structure:	Exchangeable	Num. iterations:	10
Date:	Mon, 08 Dec 2025	Scale:	1.000
Covariance type:	robust	Time:	21:08:04

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-1.9655	0.358	-5.488	0.000	-2.667	-1.264
C(Spring_Summer_dev)[T.Overwegend matig]	0.3699	0.261	1.420	0.156	-0.141	0.881
C(Spring_Summer_dev)[T.Overwegend slecht]	0.4580	0.376	1.220	0.223	-0.278	1.194
honingdracht_2024_summer_cat[T.Average]	-0.4069	0.236	-1.723	0.085	-0.870	0.056
honingdracht_2024_summer_cat[T.Poor]	0.0683	0.362	0.189	0.850	-0.641	0.778
honingdracht_2024_autumn_cat[T.Average]	0.5175	0.274	1.892	0.059	-0.019	1.054
honingdracht_2024_autumn_cat[T.Poor]	0.7746	0.317	2.440	0.015	0.152	1.397
C(Varroa_Severity)[T.Disease_varroa_Mild]	-0.4914	0.245	-2.007	0.045	-0.971	-0.012
C(Varroa_Severity)[T.Disease_varroa_Severe]	-1.1255	0.629	-1.789	0.074	-2.358	0.108
C(Varroa_Severity)[T.Disease_deformed_wings]	-0.1485	0.393	-0.378	0.706	-0.919	0.622
C(Winter_honey)[T.Deels]	1.0213	0.260	3.922	0.000	0.511	1.532
C(Winter_honey)[T.Ja]	1.3697	0.322	4.258	0.000	0.739	2.000
C(Varroa_treatment)[T.Deels]	-0.7078	0.413	-1.712	0.087	-1.518	0.103
C(Varroa_treatment)[T.Nee]	0.5059	0.321	1.578	0.114	-0.122	1.134
C(Beekeeping_method)[T.B]	-0.4788	0.409	-1.172	0.241	-1.280	0.322
C(Beekeeping_method)[T.C]	-0.1532	0.279	-0.550	0.583	-0.699	0.393

Skew:	1.0605	Kurtosis:	-0.4042
Centered skew:	0.7041	Centered kurtosis:	0.9095

Rows in red: $p<0.05$
These variables significantly associate with higher hive mortality.

Multi-Variable Analysis

GEE Regression Results

Dep. Variable:	Hive_Status	No. Observations:	1516
Model:	GEE	No. clusters:	187
Method:	Generalized Estimating Equations	Min. cluster size:	1
		Max. cluster size:	80
Family:	Binomial	Mean cluster size:	8.1
Dependence structure:	Exchangeable	Num. iterations:	10
Date:	Mon, 08 Dec 2025	Scale:	1.000
Covariance type:	robust	Time:	21:08:04

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-1.9655	0.358	-5.488	0.000	-2.667	-1.264
C(Spring_Summer_dev)[T.Overwegend matig]	0.3699	0.261	1.420	0.156	-0.141	0.881
C(Spring_Summer_dev)[T.Overwegend slecht]	0.4580	0.376	1.220	0.223	-0.278	1.194
honingdracht_2024_summer_cat[T.Average]	-0.4069	0.236	-1.723	0.085	-0.870	0.056
honingdracht_2024_summer_cat[T.Poor]	0.0683	0.362	0.189	0.850	-0.641	0.778
honingdracht_2024_autumn_cat[T.Average]	0.5175	0.274	1.892	0.059	-0.019	1.054
honingdracht_2024_autumn_cat[T.Poor]	0.7746	0.317	2.440	0.015	0.152	1.397
C(Varroa_Severity)[T.Disease_varroa_Mild]	-0.4914	0.245	-2.007	0.045	-0.971	-0.012
C(Varroa_Severity)[T.Disease_varroa_Severe]	-1.1255	0.629	-1.789	0.074	-2.358	0.108
C(Varroa_Severity)[T.Disease_deformed_wings]	-0.1485	0.393	-0.378	0.706	-0.919	0.622
C(Winter_honey)[T.Deels]	1.0213	0.260	3.922	0.000	0.511	1.532
C(Winter_honey)[T.Ja]	1.3697	0.322	4.258	0.000	0.739	2.000
C(Varroa_treatment)[T.Deels]	-0.7078	0.413	-1.712	0.087	-1.518	0.103
C(Varroa_treatment)[T.Nee]	0.5059	0.321	1.578	0.114	-0.122	1.134
C(Beekeeping_method)[T.B]	-0.4788	0.409	-1.172	0.241	-1.280	0.322
C(Beekeeping_method)[T.C]	-0.1532	0.279	-0.550	0.583	-0.699	0.393

Skew:	1.0605	Kurtosis:	-0.4042
Centered skew:	0.7041	Centered kurtosis:	0.9095

Rows in green: 0.05< p <0.1.
These variables associate with higher hive mortality with less significance.

Conclusion

- **1. Sugar vs Honey: Not using honey effectively reduces death**
- Hives fed fully with honey were 4 times more likely to die compared to hives fed with sugar. Even mixing honey with sugar increased the risk by nearly 3 times.
- **2. Varroa and Treatment: Not observing doesn't mean not existing**
- Even when beekeepers reported 'not observing varroa infection', those who don't treat varroa has a significantly higher death rate than those who did.
- Treatment for varroa works. The biggest risk is assuming hives are varroa-free and do nothing.
- **3. Quality of honey flow during autumn is a sign for survival**
- The hives with a poor rating on honey flow is 2 times more likely to die than other hives.
- **4. Beekeeping Method itself doesn't matter**
- Statistical analysis showed no significant difference in survival rates between Biodynamic and more controlled methods. It's not directly about the 'philosophy' behind beekeeping itself, but what beekeepers actually do that influences mortality.

Appendix

Reference

- Vervaardiging en nauwkeurigheid van het LGN3-grondgebruiksbestand.
- Wit, A.J.W. de, Heijden, Th.G.C. van der en H.A.M. Thunnissen, 1999. Wageningen, DLO-Staring Centrum, Rapport 663.
- Landelijk Grondgebruiksbestand Nederland (LGN5). Vervaardiging, nauwkeurigheid en gebruik.
- Hazeu, G.W., 2005. Wageningen, Alterra. Alterra-report 1213, 92p., 18 figs., 11 tables and 11 refs. DOI: https://doi.org/10.18174/17654
- Landelijk Grondgebruiksbestand Nederland versie 6 (LGN6). Vervaardiging, nauwkeurigheid en gebruik.
- Hazeu, G.W., Schuiling, C., Dorland, G.J., Oldengarm, J. & H.A. Gijsbertse, 2010. Wageningen, Alterra. Alterra-rapport 2012, 132p., 20 figs., 9 tables and 15 refs. DOI: https://doi.org/10.18174/137531

- Hazeu, G., R. Schuiling, D. Thomas, M. Vittek, M. Storm en J.D. Bulens, 2023. Wageningen, Wageningen Environmental Research, Rapport 3235. 100 blz.; 10 fig.; 8 tab.; 16 ref. https://lgn.nl/reports/WENR-rapport%203235_Totaal_LR.pdf#page=59.11'
- Ctgb. Guidelines for maximum residue levels in honey. https://english.ctgb.nl/latest/news/2019/05/06/guidelines-for-maximum-residue-levels-in-honey
- Technical guidelines for determining the magnitude of pesticide residues in honey and setting Maximum Residue Levels in honey. https://food.ec.europa.eu/document/download/06bcc245-57b5-47ff-b31b-25c05a8485cf_en?filename=pesticides_mrl_guidelines_honey.pdf

Single Variable Hypothesis Test

- Unweighted, beekeeper level

Variable Name	Test Method	P-value	Sig.
Dev_status_summer_24	Kruskal-Wallis	0.0139	*
Varroa_severity	Kruskal-Wallis	0.0214	*
Disease_aziatisch_hoornaar	Mann-Whitney	0.7526	
honingdracht_2024_autumn	Spearman Corr	0.0044	** (r=-0.21)
honingdracht_2024_summer	Spearman Corr	0.0059	** (r=-0.20)

Spring and Summer Development * Survival Rate

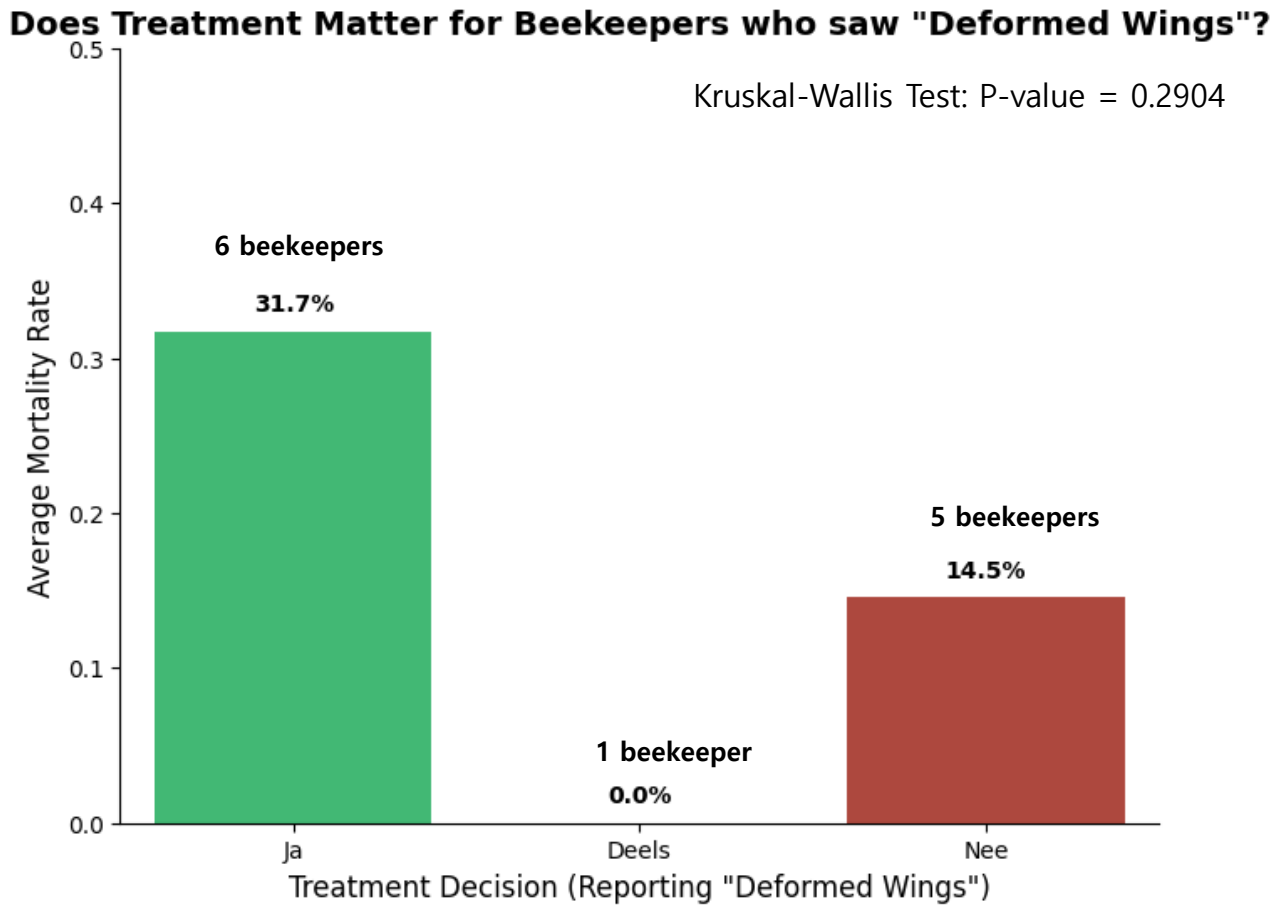
- Unweighted, beekeeper level
- Median survival rate is 100% for both 'good' and 'moderate' groups.

```
--- Median Survival Rates ---
Dev_status_summer_24
A    1.00
B    1.00
C    0.65
Name: Survival_Rate, dtype: float64
                       count      mean       std  min       25%   50%  \
Dev_status_summer_24
A                      139.0  0.828714  0.302276  0.0  0.683333  1.00
B                       34.0  0.725770  0.364517  0.0  0.500000  1.00
C                       11.0  0.612121  0.316587  0.0  0.500000  0.65

                            75%  max
Dev_status_summer_24
A                      1.000000  1.0
B                      1.000000  1.0
C                      0.833333  1.0
```

Treatment and Self-Reported Disease

- Unweighted, beekeeper level
- The number of beekeepers belong to this category is too few.



Death Rate Comparison: Unweighted and weighted

- The average death rate by varroa severity differs greatly on beekeeper level (unweighted) and on hive population level (weighted).

	Unweighted	Weighted
Varroa_severity	Beekeeper_Avg	Population_Avg
None	0.278818	0.202797
Disease_varroa_Mild	0.121396	0.078014
Disease_varroa_Severe	0.109524	0.048387
Disease_deformed_wings	0.218939	0.220588

Categorization +Hypothesis Test for Kasttype

- Weighted, hive level

Label	Category	Kast
A	**Controlled**	Dadant, Segeberger, Spaarkast, Simplexkast, WBC, DuHoKa, Zesramer
B	**Natural**	TBH, Topbarhive, Topkast, Natuurbouwkast, Korf, Hangkorf, Warre
C	**Mix of A and B**	

```
==================== Variable: Kasttype ====================
Contingency Table (Observed):
Status       0    1
Kasttype
A         1213  191
B           36    6
C           93   16

Minimum expected frequency: 5.75
✅ Pearson Chi-Square Test
Chi2 Statistic: 0.1115, P-value: 9.4578e-01
Conclusion: Not significant. No association found between Kasttype and hive mortality.
```

Categorization +Hypothesis Test for Ras

- Weighted, hive level

```
==================== Variable: Ras ====================
Contingency Table (Observed):
Status      0   1
Ras
Buckfast  339  68
Carnica   495  85
Geen      432  73
Zwarte     59   5

Minimum expected frequency: 9.50
✅ Pearson Chi-Square Test
Chi2 Statistic: 3.6978, P-value: 2.9599e-01
Conclusion: Not significant. No association found between Ras and hive mortality.
```

Multi-Variable Analysis

- Honeyflow as numerical
- Result is similar to the GEE model where honeyflow as ordinal categorical.

	Feature	VIF
0	const	33.212902
13	honingdracht_2024_autumn	1.762913
12	honingdracht_2024_summer	1.620143
6	Winter_honey_Deels	1.429695
7	Winter_honey_Ja	1.420100
10	Beekeeping_method_B	1.405946
3	Varroa_Severity_Disease_varroa_Mild	1.317993
8	Varroa_treatment_Deels	1.304311
9	Varroa_treatment_Nee	1.302175
4	Varroa_Severity_Disease_varroa_Severe	1.229129
5	Varroa_Severity_Disease_deformed_wings	1.200957
11	Beekeeping_method_C	1.196586
1	Spring_Summer_dev_Overwegend matig	1.179896
2	Spring_Summer_dev_Overwegend slecht	1.150796

GEE Regression Results

Dep. Variable:	Hive_Status	No. Observations:	1516
Model:	GEE	No. clusters:	187
Method:	Generalized Estimating Equations	Min. cluster size:	1
		Max. cluster size:	80
Family:	Binomial	Mean cluster size:	8.1
Dependence structure:	Exchangeable	Num. iterations:	9
Date:	Mon, 08 Dec 2025	Scale:	1.000
Covariance type:	robust	Time:	21:07:17

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-0.7517	0.497	-1.514	0.130	-1.725	0.222
C(Spring_Summer_dev, Treatment('Overwegend goed'))[T.Overwegend matig]	0.1966	0.256	0.768	0.442	-0.305	0.698
C(Spring_Summer_dev, Treatment('Overwegend goed'))[T.Overwegend slecht]	0.4350	0.404	1.077	0.282	-0.357	1.227
C(Varroa_Severity, Treatment('None'))[T.Disease_varroa_Mild]	-0.4775	0.243	-1.969	0.049	-0.953	-0.002
C(Varroa_Severity, Treatment('None'))[T.Disease_varroa_Severe]	-1.1251	0.597	-1.885	0.059	-2.295	0.045
C(Varroa_Severity, Treatment('None'))[T.Disease_deformed_wings]	-0.0368	0.378	-0.097	0.923	-0.778	0.704
C(Winter_honey, Treatment('Nee'))[T.Deels]	1.0174	0.267	3.804	0.000	0.493	1.542
C(Winter_honey, Treatment('Nee'))[T.Ja]	1.4197	0.321	4.429	0.000	0.791	2.048
C(Varroa_treatment, Treatment('Ja'))[T.Deels]	-0.6578	0.412	-1.598	0.110	-1.465	0.149
C(Varroa_treatment, Treatment('Ja'))[T.Nee]	0.4481	0.327	1.369	0.171	-0.194	1.090
C(Beekeeping_method, Treatment('A'))[T.B]	-0.5124	0.412	-1.243	0.214	-1.320	0.296
C(Beekeeping_method, Treatment('A'))[T.C]	-0.1755	0.281	-0.625	0.532	-0.726	0.375
honingdracht_2024_summer	0.0018	0.064	0.028	0.978	-0.123	0.126
honingdracht_2024_autumn	-0.1537	0.057	-2.684	0.007	-0.266	-0.041

Skew:	1.0729	Kurtosis:	-0.3861
Centered skew:	0.7041	Centered kurtosis:	0.9095

Multicollinearity Test

- VIF < 5: No multicollinearity

	Feature	VIF
0	const	11.901838
15	honingdracht_2024_autumn_cat_Poor	2.460638
14	honingdracht_2024_autumn_cat_Average	1.695748
6	Winter_honey_Deels	1.549580
13	honingdracht_2024_summer_cat_Poor	1.540046
7	Winter_honey_Ja	1.466731
12	honingdracht_2024_summer_cat_Average	1.440837
8	Varroa_treatment_Deels	1.403108
10	Beekeeping_method_B	1.392374
3	Varroa_Severity_Disease_varroa_Mild	1.386962
9	Varroa_treatment_Nee	1.340251
4	Varroa_Severity_Disease_varroa_Severe	1.262112
5	Varroa_Severity_Disease_deformed_wings	1.242432
2	Spring_Summer_dev_Overwegend slecht	1.198256
11	Beekeeping_method_C	1.196079
1	Spring_Summer_dev_Overwegend matig	1.137796